

Figure X. Calibration vs stress overall\_score across three KEGG-curated datasets.

v1 = default thresholds (binary mean\_completeness  $\geq 50\%$ ); v2 = dominance-aware (v0.9.x min\_dominance\_fraction = 0.20).

Cross-topic arsenate\_reduction rejected with n=0 active MAGs in 2/2 non-arsenic datasets ★.

